A Food Waste Rescue Marketplace and Demand Forecasting Application

Eddie Allman, Marcos Ashton, Theo Chan, Rafe Crowley, Caspar Hadley, Ben Parker, Charlie Winders, Jason Yeung

1. Executive Summary

1.1 Problem Identified

Using data from 2021 and 2022, it has been estimated that 11% of the UK's total food waste has been created by the hospitality and food service industry[3]. Globally, food loss and waste accounts for 8-10% of greenhouse gas emissions[1], meaning that hospitality food waste is directly responsible for up to 1.1% of total greenhouse pollution. Shifting focus to employee statistics, a survey on staff productivity has shown that over an average 8 hour working day, only 2 hours and 53 minutes are spent productively[2]. Furthermore, this study shows that most workers get takeway etc after a long day costing an average xxx pounds per day.

1.2 Proposed Solution

2. Prioritised Requirements

An explanation of the features we think could possible be included in the final version of Byte Me, and justification for those that are being implemented in Sprint 1. Features are classified in accordance with the MoSCoW format; Must have, should have, could have, and will not have.

2.1 Prioritisation Rationale

The prioritisation strategy focuses on delivering a reliable vertical slice in Sprint 1 that satisfies all mandatory coursework requirements while clearly demonstrating system value. Core marketplace functionality (posting, reserving, claim code verification, and streak updates), baseline analytics, forecasting, and a fully seeded dataset are treated as non-negotiable because they directly map to the marking criteria and defined measures of success (zero oversell rate, functioning forecasting evaluation, and measurable waste impact). Sprint 2 expands depth rather than breadth, strengthening analytics, forecasting accuracy, governance documentation, and maintainability to support client handover and professional practice evidence. Enhancements such as maps, reviews, and extended gamification are deliberately deprioritised to avoid scope creep, while payment processing and gambling-style mechanics are explicitly excluded due to specification constraints and legal risk. This structured approach ensures technical completeness, risk control, and alignment with both CW1 and CW2 assessment objectives.

2.2 MoSCoW Prioritisation

Must Have (Sprint 1)	**Should Have (Sprint 2)**	**Could Have (If Time)**	**Will Not Have (Out of Scope)**
Login and role-based access control	UX polish and accessibility refinement	Advanced dietary preference filters	In-app payment processing (explicitly disallowed)
Clean landing page	Map view of nearby stores	Seller and buyer review system	Paid spin / gambling-style rewards
Seller bundle posting (quantity, price, pickup window, allergens)	Issue reporting workflow	Google Maps walking-time integration	Real-time delivery tracking
Consumer browsing and reservation system	Terms of Service page	"Meet the Developers" page	Live chat support
Claim code generation and verification	Cookies policy (if applicable)	Extended gamification features	Full e-commerce checkout system
Rescue streak tracking (mandatory requirement)	Data protection and privacy policy	Advanced seller dashboard customisation	Social media integration
Minimum seller analytics (sell-through and waste proxy)	Full analytics suite (pricing and operational insights)	Leaderboard functionality	Complex loyalty marketplace
Baseline forecasting model and evaluation snapshot	Improved forecasting model with recommendation logic	Weather API integration	AI chatbot assistant
Seeded dataset (25+ sellers, 250+ bundles, 400+ reservations)	Maintainer/admin tools	Extended accessibility enhancements	Real payment gateway integration
Allergen display and safety disclaimer	Formal accessibility audit		

Table 2.1: Expanded MoSCoW prioritisation aligned to coursework specification.

2.2.1 Sprint Allocation and Justification

Feature	**Sprint**	**Justification**
End-to-end marketplace workflow (Post → Reserve → Claim → Streak update)	Sprint 1	Required vertical slice for CW1 demonstration. Core system value depends on reliable end-to-end execution.
Seller analytics (minimum two views)	Sprint 1	Explicit CW1 requirement demonstrating measurable waste reduction and operational insight.
Baseline forecasting model	Sprint 1	Specification requires at least one baseline forecast and evaluation snapshot in CW1.
Seeded dataset	Sprint 1	Mandatory for marking; system must operate without external services.
Issue reporting system	Sprint 2	Required for full v1.0 system but not part of Sprint 1 vertical slice.
Expanded analytics suite	Sprint 2	CW2 requires deeper analytical evaluation and pricing effectiveness insights.
Improved forecasting with recommendation logic	Sprint 2	CW2 requires comparison against baselines on held-out data and actionable recommendations with confidence labels.
Maintainer tools and deployment documentation	Sprint 2	Part of the client handover pack and professional practice evidence for final submission.
Accessibility refinement	Sprint 2	Baseline compliance achieved in Sprint 1; full audit and refinement in Sprint 2.
Policies (ToS, Privacy, Cookies)	Sprint 1 / 2	Basic policy links included in Sprint 1; expanded documentation aligned with final system in Sprint 2.
Maps, reviews, extended gamification	If time (Sprint 2+)	Enhancement features that do not affect system reliability, forecasting accuracy, or core marking criteria.
Payments and gambling-style mechanics	Out of Scope	Explicitly prohibited by coursework brief and introduce unnecessary legal and security risk.

Table 2.2: Sprint allocation and rationale aligned to CW1 and CW2 marking criteria.

3. Architecture v1

Diagrams and explanations from the design phases of our agile development. Included is information about our frontend, backend, and database systems, along with data flow and user interaction specifics.

3.1 Sitemap and User Journey

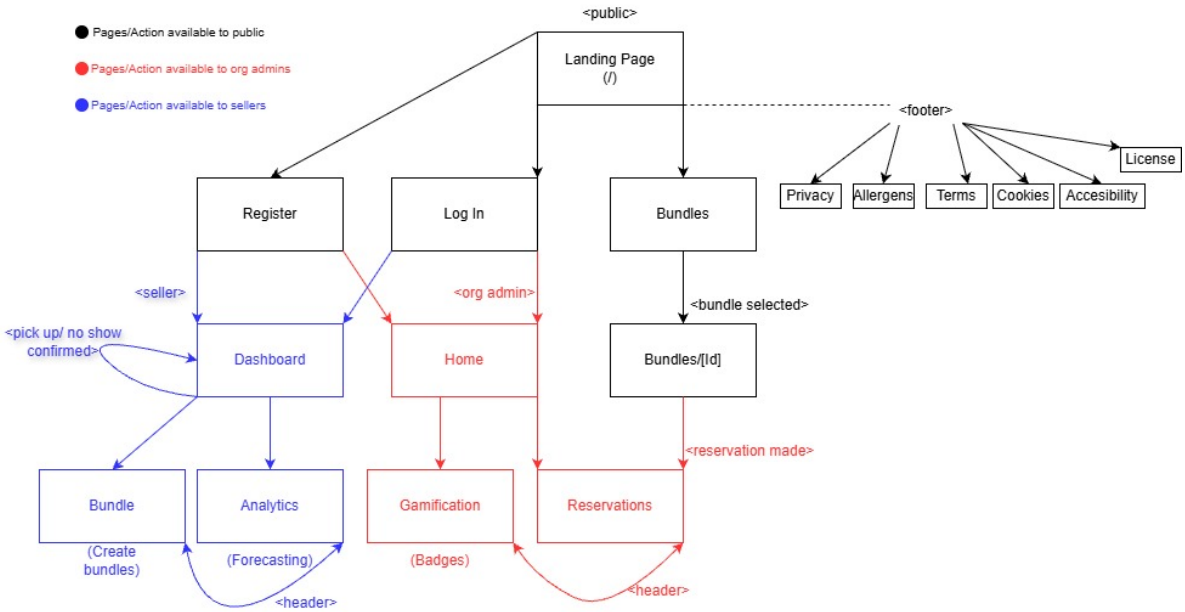


Figure 3.1: User Flow Diagram for Byte Me

3.1.1 User Flow Justification

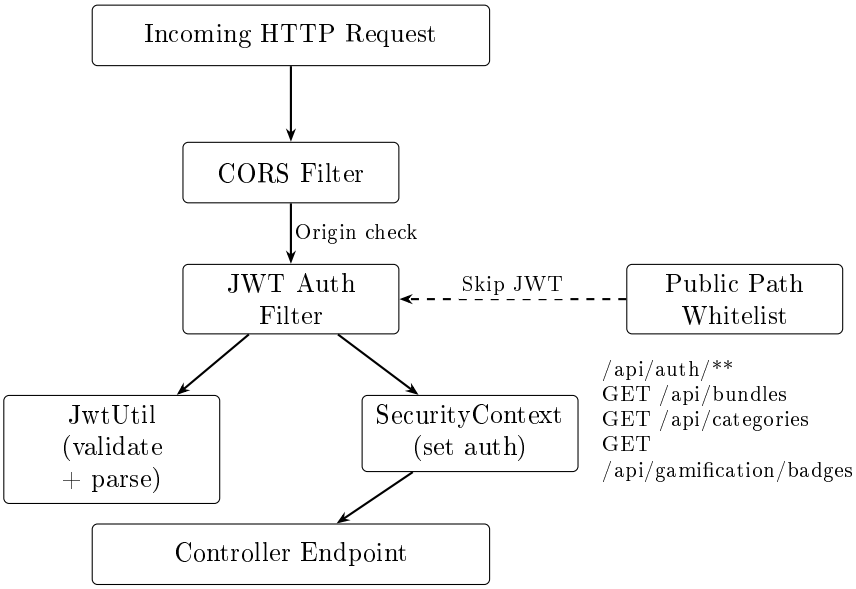
Public → When a user initially loads the website thy see our landing page. This displays key details about how the project works and its goals, as well as giving them access to log in and register pages, alongside key pages via. links in the header and footer. Without being logged in, a user can access all legality associated pages through the footer, specifically; the privacy policy, terms and conditions, cookies policy, allergens notice, accessibility statement, and license pages. It is important these remain publicly available for full disclosure to users even before they have made or logged into an account. All users are also given access to the bundles page, and through this the bundles/[id] page, giving them key details, dates and price points of any available or soon to be available bundles from our sellers.

Organisation Admins → Once an org admin has logged in or registered via. the pages on our header, they will gain authorisation to all org admin specific pages. For now, this is the gamification page, and the reservations page. They will also gain access to new features within the bundles/[id] page allowing them to reserve bundles under their organisation name and acquire a claim code for picking up said bundle. Any reservations made can be re-accessed by visiting the reservations page which is made available to org admins in the header. This allows users to reserve bundles in advance and return later to cancel or retrieve their claim code, giving our users more freedom and ease when using our service. The gamification page displays any badges achieved and any progress towards over achievable badges. This allows companies to see the important steps they are making to help the environment and reduce food waste, while giving

them real, certified badges they can display to show the work they are doing to others.

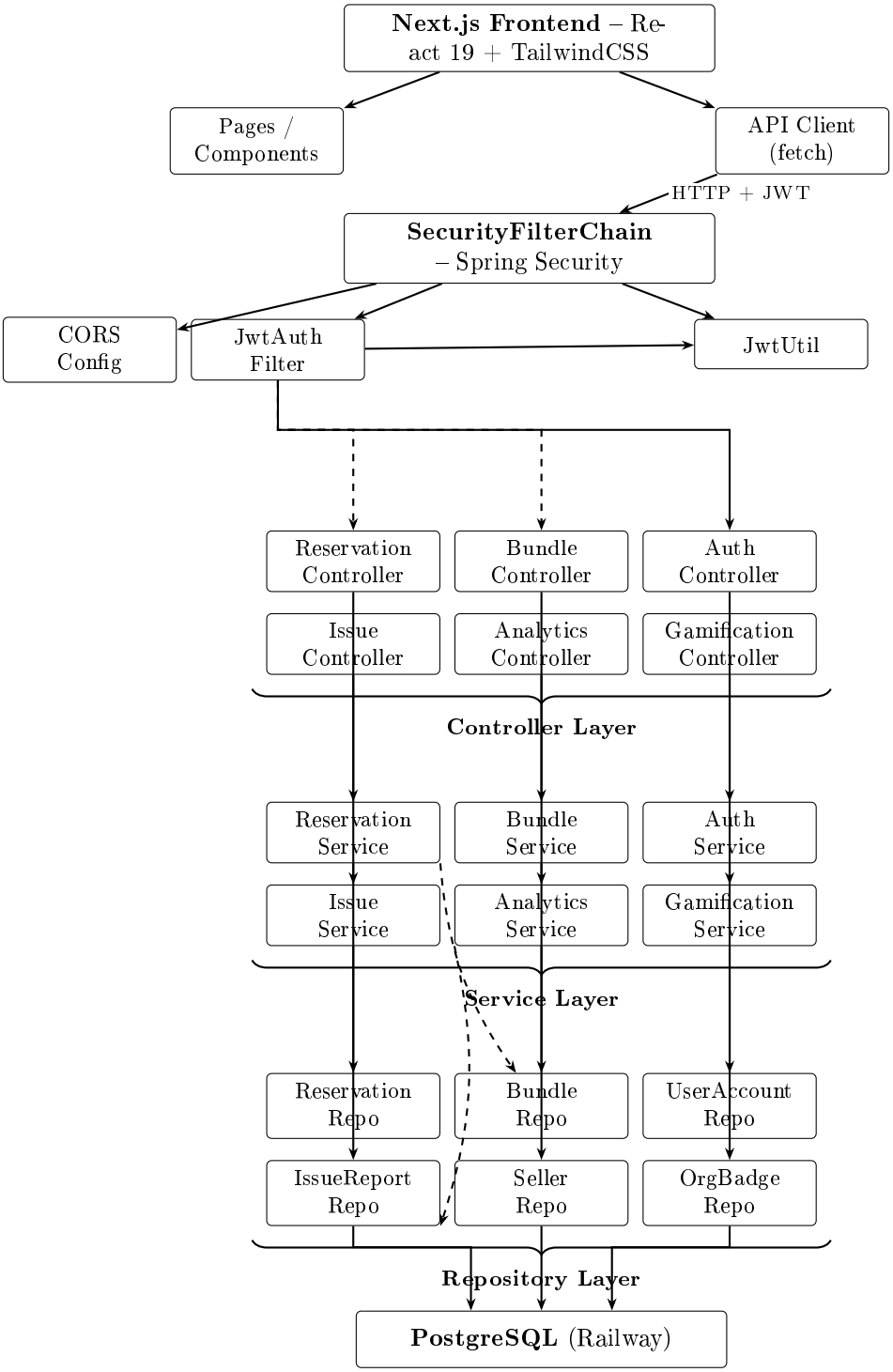
Sellers → Once logged in as a seller, the user is redirected to their personalised seller dashboard page. This gives them insight into how many bundles have been posted, sold and cancelled, alongside their active bundles and links to seller specific pages. Sellers can access a special bundle page, providing a form for them to create and post bundles which are made available immediately or at a scheduled time to all users on our platform. We believe it is important for our sellers to have a simple UI for bundle creation as we understand this is not the main focus of their business, and the ease of creation increases the amount of bundles made, and therefore reducing food wasted. Sellers can also access the analytics page, giving detailed forecasting of their demand in general, and for individual bundles. This helps them tweak supply to further minimise food waste, as well as saving money for their businesses, providing a real incentive for our sellers to waste as little food as possible.

3.2 Security Integration



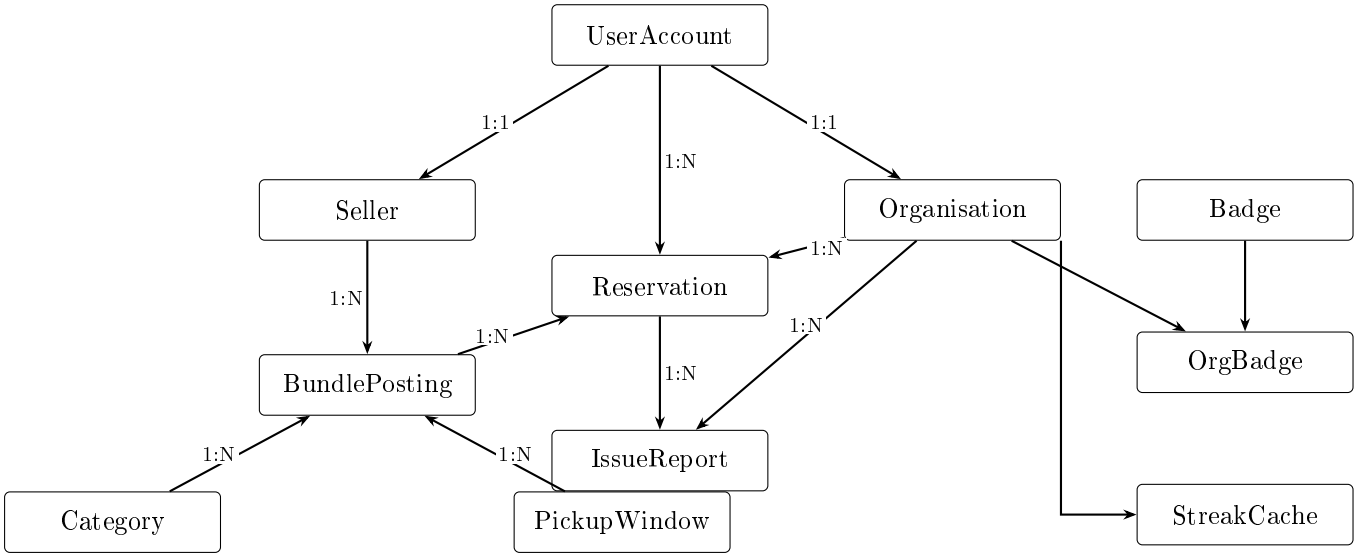
CORS runs first. JWT filter validates tokens for non whitelisted paths and sets the SecurityContext. Sessions are stateless passwords use BCrypt.

3.3 Full-Stack Integration Diagram



Frontend calls the backend via REST with JWT. Each controller delegates to its service, which queries repositories. Dashed arrows show cross-service repo access.

3.4 Entity Relationship Integration



UserAccount maps 1:1 to Seller or Organisation by role. Sellers create bundles; organisations reserve and file issues. OrganisationBadge is a many to many join. Category and PickupWindow are lookup tables.

3.5 title

4. Forecasting Plan

A summary of the forecasting techniques used within Sprint 1 of the project described within the context of the final forecasting feature's design. Covers both current and future forecasting models.

5. Success Criteria

An outline of the five key measures of success used to evaluate the Byte Me food rescue platform. Each measure is defined with specific criteria and supported by evidence from the system's codebase, ensuring reliability, accuracy, and environmental impact.

5.1 Marketplace Reliability

5.1.1 Oversell Rate (Target: 0%)

Definition: The system must never allow more reservations than the available quantity of a bundle.

Evidence: The `BundlePosting` entity enforces a strict logic check before any reservation is permitted. The `canReserve` method explicitly verifies inventory levels.

```java
    // Check if can reserve
    public boolean canReserve(int qty) {
      return status == Status.ACTIVE && (quantityReserved + qty) <= quantityTotal;
    }

```

Listing 5.1: BundlePosting.java Inventory Logic

Furthermore, the `OrderController` transactionally locks this process. If `canReserve(1)` returns false, the system immediately rejects the request with a `400 Bad Request`, ensuring the oversell rate remains strictly at 0.

5.1.2 Handling Expiries and No-Shows

Definition: The lifecycle of every bundle must be accounted for, distinguishing between successful collections, expired goods, and user no-shows.

Evidence:

- **Status Tracking:** The `Reservation.Status` enum includes specific states for `NO_SHOW`, `EXPIRED`, and `CANCELLED`.
- **Automated Cleanup:** The `BundlePostingRepository` includes a query to identify expired bundles automatically:

```java
    @Query("SELECT b FROM BundlePosting b WHERE b.status = 'ACTIVE' AND b.pickupEndAt < :now")
    List<BundlePosting> findExpired(Instant now);

```

Listing 5.2: Finding Expired Bundles

This ensures that inventory lists remain fresh and accurate without manual intervention.

5.2 Forecast Accuracy

5.2.1 Methodology

We compare our chosen predictive model against standard baselines using a Held-Out Evaluation approach (80% training data, 20% evaluation data).

5.2.2 Metrics & Baselines

The `ForecastService` calculates three key error metrics for every model run:

1. **MAE (Mean Absolute Error):** Average magnitude of reservation errors.
2. **RMSE (Root Mean Squared Error):** Penalizes larger outliers.
3. **Brier Score:** Evaluates the accuracy of probabilistic no-show predictions.

These metrics are calculated by comparing the **Chosen Model (EMA-Weighted)** against two baselines:

- **Moving Average (4-Week):** A simple sliding window average.
- **Seasonal Naive:** Predicts based on the same ISO Day of Week from history.

```
// Compare models and save metrics
Map<String, Object> maMetrics = Map.of(
"MAE_reservations", calculateMAE(maPredicted, actualRes),
"RMSE_reservations", calculateRMSE(maPredicted, actualRes),
"Brier_no_show", calculateBrierScore(maNoShowPred, actualNoShow));

```

Listing 5.3: Model Comparison Logic

The system stores these results in the `ForecastRun` entity to allow for continuous performance monitoring.

5.3 Waste Impact & Carbon Reduction

5.3.1 Estimated Waste Avoided

Measure: The total weight of food diverted from landfills.

Evidence: The `SellerMetricsWeekly` entity tracks `wasteAvoidedGrams`. This is calculated by multiplying the number of `COLLECTED` reservations by the `estimatedWeightGrams` defined in the bundle posting.

5.3.2 Real-World Example: CO2e Prevention

To illustrate, consider the sample data for "Sourdough & Co Bakery" used in our forecast demo.

- **Scenario:** A bakery posts surplus bundles with an estimated weight of 800g each.
- **Observations:** On Jan 30, 2026, the bakery observed 16 reservations with a 9% no-show rate.
- **Food Waste Avoided:**

$$\text{Mass} = 16 \times (1 - 0.09) \times 0.8 \text{ kg} \approx 11.65 \text{ kg}$$

- **Carbon Impact:** Using a standard conversion factor of 2.5 kg CO2e per kg of food waste prevented (WRAP, 2023):

$$\text{CO2e Saved} = 11.65 \text{ kg} \times 2.5 = \mathbf{29.1 \text{ kg CO2e}}$$

This metric is aggregated weekly and displayed on the seller dashboard, providing tangible feedback on their environmental contribution. The `CO2_SAVER` badge is explicitly awarded when organizations reach 10 collections, further reinforcing this goal.

5.4 User Experience (Task-Based)

5.4.1 Reserving and Claiming

Target: Sub-second response times for reservation and collection tasks.

Evidence:

- **Instant Reservation:** The `create` order endpoint generates a secure 6-digit `claimCode` immediately upon success.
- **Secure Collection:** The collection process uses a hashed verification system. When an organisation presents a code, the backend verifies it using `passwordEncoder.matches()`.

```
    if (!passwordEncoder.matches(claimReq.getClaimCode(), reservation.
    getClaimCodeHash())) {
      return ResponseEntity.badRequest().body("Invalid claim code");
    }

```

Listing 5.4: Claim Code Verification

This approach balances high security with the speed required for retail environments.

5.5 Gamification & Engagement

5.5.1 Measures

- **Retention:** Tracking `currentStreakWeeks` via the `OrganisationStreakCache`.
- **Milestones:** Awarding badges like `STREAK_4` and `CO2_SAVER`.

Evidence: The system automatically updates streak counters and checks for badge eligibility after every successful collection, providing immediate positive reinforcement to the user.

6. Sprint 2 Plan

Features we intend to implement in the second development phase of Byte Me and their timelines.

Bibliography

[1] FAO, F. A. A. O. O. T. U. N. FAO Policy Series: Food Loss & Food Waste, 2013.

[2] LTD, I. D. Survey Reveals Employee Productivity Averages 2 Hours and 53 Minutes a Day, Jan. 2025.

[3] WRAP. UK Food Waste & Food Surplus - Key Facts. Tech. rep., July 2025.